

**INSTITUTE OF INFORMATION TECHNOLOGY
UNIVERSITY OF DHAKA**



<http://www.iit.du.ac.bd/>

**SYLLABUS FOR
BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING (BSSE)
Effective from Session 2022-23**

Semester-Wise Courses

Semester 1

<i>Course Code</i>	<i>Course Title</i>	<i>Credit</i>	<i>Theory</i>	<i>Lab</i>
CSE 1101	Structured Programming	3	3	0
CSE 1102	Discrete Mathematics	3	3	0
STAT 1103	Probability and Statistics for Engineers I	3	3	0
MATH 1107	Calculus I	3	3	0
GE 1105	Introduction to Sociology	3	3	0
SE 1106	Introduction to Software Engineering	3	3	0
CSE 1101L	Structured Programming Lab	1.5	0	1.5
Total Credits		19.5	18	1.5

Semester 2

<i>Course Code</i>	<i>Course Title</i>	<i>Credit</i>	<i>Theory</i>	<i>Lab</i>
CSE 1215	Data Structure	1.5	1.5	0
MATH 1216	Calculus II	3	3	0
CSE 1211	Computer Organization	3	3	0
STAT 1203	Probability and Statistics for Engineers II	3	3	0
GE 1212	Bangladesh Studies	3	3	0
SE 1206	Object Oriented Concepts	3	3	0
CSE 1215L	Data Structure Lab	1.5	0	1.5
CSE 1211L	Computer Organization Lab	0.75	0	0.75
SE 1206L	Object Oriented Concepts Lab	1.5	0	1.5
Total Credits		20.25	16.5	3.75

Semester 3

<i>Course Code</i>	<i>Course Title</i>	<i>Credit</i>	<i>Theory</i>	<i>Lab</i>
CSE 2115	Introduction to Algorithm	3	3	0
SE 2112	Theory of Computing	3	3	0
CSE 2111	Computer Networking	3	3	0
MATH 2116	Linear Algebra	3	3	0
SE 2105	Software Project Lab I	3	0	3
BUS 2117	Business Communication	3	3	0
CSE 2115L	Introduction to Algorithm Lab	1.5	0	1.5
CSE 2111L	Computer Networking Lab	0.75	0	0.75
Total Credits		20.25	15	5.25

Semester 4

<i>Course Code</i>	<i>Course Title</i>	<i>Credit</i>	<i>Theory</i>	<i>Lab</i>
CSE 2201	Operating System and System Programming	3	3	0
SE 2215	Design Patterns	1.5	1.5	0
CSE 2204	Database Management System	3	3	0
BUS 2205	Business Studies for Engineers	3	3	0
SE 2206	Software Requirements Specification and Analysis	1.5	1.5	0
CSE 2201L	Operating System and System Programming Lab	1.5	0	1.5
SE 2215L	Design Patterns Lab	1.5	0	1.5
CSE 2204L	Database Management System Lab	1.5	0	1.5
SE 2206L	Software Requirements Specification and Analysis Lab	1.5	0	1.5
Total Credits		18	12	6

Semester 5

<i>Course Code</i>	<i>Course Title</i>	<i>Credit</i>	<i>Theory</i>	<i>Lab</i>
SE 3115	Software Testing and Quality Assurance	1.5	1.5	0
CSE 3102	Web Technology	1.5	1.5	0
CSE 3116	Mathematical and Statistical Methods for Data Analysis	3	3	0
CSE 3117	Artificial Intelligence	3	3	0
SE 3118	Software Design and Analysis	1.5	1.5	0
CSE 3119	Cryptography and Security Mechanisms	3	3	0
SE 3105	Software Project Lab II	3	0	3
CSE 3102L	Web Technology Lab	1.5	0	1.5
CSE 3119L	Cryptography and Security Mechanisms Lab	0.75	0	0.75
Total Credits		18.75	13.5	5.25

Semester 6

<i>Course Code</i>	<i>Course Title</i>	<i>Credit</i>	<i>Theory</i>	<i>Lab</i>
SE 3215	Internship	15	0	15
Total Credits		15	0	15

Semester 7

<i>Course Code</i>	<i>Course Title</i>	<i>Credit</i>	<i>Theory</i>	<i>Lab</i>
CSE 4115	Distributed Systems	1.5	1.5	0
CSE 4116	Compiler Design	3	3	0
SE 4117	IT Laws & Ethics	3	3	0
CSE/SE 41XX	Elective 1	3	3	0
CSE 4118	Machine Learning	3	3	0
CSE 4115L	Distributed Systems Lab	1.5	0	1.5
SE 4119	Project	3	0	3
Total Credits		18	13.5	4.5

Semester 8

<i>Course Code</i>	<i>Course Title</i>	<i>Credit</i>	<i>Theory</i>	<i>Lab</i>
SE 4215	Software Security	1.5	1.5	0
SE 4203	Software Project Management	3	3	0
CSE/SE 42XX	Elective 2	3	3	0
CSE/SE 42XX	Elective 3	3	3	0
SE 4201	Project	3	0	3
SE 4215L	Software Security Lab	1.5	0	1.5
SE 4216	Software Metrics	1.5	1.5	0
SE 4216L	Software Metrics Lab	0.75	0	0.75
Total Credits		17.25	12	5.25

Elective Courses

<i>No.</i>	<i>Track</i>	<i>Course Code</i>	<i>Course Title</i>
1	Data Science	CSE 4150	Information Retrieval
		CSE 4250 CSE 4251	Natural Language Processing Introduction to Data Mining
2	Security	CSE 4160	Network and System Security
		CSE 4261 CSE 4262	Digital Forensics Cyber Physical System Security
3	Computer Vision and Image Processing	CSE 4170	Pattern Recognition
		CSE 4270 CSE 4271	Image Processing Photogrammetry
4	Bioinformatics	CSE 4280 CSE 4281	Introduction to Bioinformatics Genomic Data Science
5	Advanced Software Engineering	SE 4290 SE 4291 SE 4292	Software Maintenance Human Computer Interaction Service-Oriented Architecture

Semester 1 (1st year 1st Semester)

Course No.	Course Title	No. of Credits	Credit Hours
CSE 1101	Structured Programming	Theory: 3	15 × (1×3) = 45

Course Contents:

Fundamentals of C programming; Introducing C's Program Control Statements; Data types, Variables and Expressions; Exploring Arrays and Strings; Understanding Pointers and Functions; Console and File I/O; Structures and Unions.

References:

1. Teach Yourself C, Herbert Schildt, McGraw Hill
2. C: The Complete Reference, Herbert Schildt, McGraw Hill
3. Schaum's Outline of programming with C, McGraw Hill

Course No.	Course Title	No. of Credits	Credit Hours
CSE 1102	Discrete Mathematics	Theory: 3	15 × (1×3) = 45

Course Contents:

Logic and Proofs: propositional logic, applications of propositional logic, propositional equivalences, predicates and quantifiers, nested quantifiers, rules of inference, introduction to proofs. **Induction and Recursion:** Mathematical Induction, Recursive Definitions and Structural Induction, Strong induction. **Basic Structures:** Sets, Functions, Sequences, Sums, Matrices. **Relations:** Symmetry, transitivity, reflexivity, representing relations, closure of relations, equivalence classes, partial orderings. **Graphs:** Graphs and graph models, graph terminology, representing graphs, graph isomorphism, connectivity, graph coloring, Euler and Hamilton paths. **Trees:** Introduction and application of trees, introduction to binary search tree and decision tree, tree traversal, spanning trees. **Counting:** The addition and multiplication rules, The principle of Inclusion-Exclusion, The pigeon-hole principle, permutations, combinations, Generalized Permutations and Combinations. **Number Theory:** The division algorithm, divisibility and the Euclidean algorithm, prime numbers, congruence, applications of congruence;

References:

1. Discrete Mathematics and Its Applications, 8th Ed. – Kenneth H. Rosen

2. Discrete Mathematics: An Open Introduction, 3rd Ed. – Oscar Levin
3. Discrete Mathematics with Applications, 4th Ed. – Susanna S. Epp

Course No.	Course Title	No. of Credits	Credit Hours
STAT 1103	Probability and Statistics for Engineers I	Theory: 3	15 × (1×3) = 45

Course Contents: Introduction to Statistics: Concept of Data and Variables, Data Collection and Descriptive Statistics, Inferential Statistics; Populations and Samples; Descriptive Statistics: Frequency Tables and Graphs, Relative Frequency Tables and Graphs, Grouped Data, Histograms, Ogives, Stem and Leaf Plots, Sample Mean, Sample Median, Sample Mode, Sample Variance and Standard Deviation, Sample Percentiles and Box Plots, Chebyshev's Inequality, Normal Data Sets, Paired Data Set and Sample Correlation Coefficient; Elements of Probability: Basic Terminology in Probability, Sample Space and Events, Venn Diagrams and Algebra of Events, Axioms of Probability, Conditional Probability, Bayes' Theorem and Independent Events; Random Variables and Expectation: Random Variables, Types of Random Variables, Jointly Distributed Random Variables, Expectation, Property of Expected Values, Use of Expected Values in Decision Making, Variance, Covariance and Variance of Sums of Random Variables and Moment Generating Functions; Special Random Variables: Binomial Random Variables, Poisson Random Variables, Uniform Random Variables, Normal Random Variables, Exponential Variables, Gamma Distribution, Chi-Square Distribution, t-Distribution and F-Distribution; Distributions of Sampling Statistics: Central Limit Theorem, Sampling Distribution for Normal Population, and Sampling from a Finite Population; Parameter Estimation: Maximum Likelihood Estimators, Interval Estimates, Estimating the difference in Means of Two Normal Population, Approximate Confidence Interval for the Mean, Confidence Interval of the Mean of the Exponential Distribution and Bayes' Estimator.

References:

1. Sheldon M. Ross, Introduction to Probability and Statistics for Engineers and Scientists, Elsevier/Academic Press, 4th Ed.
2. Hayter A, Probability and Statistics for Engineers and Scientists, Cengage Learning, 4th edition

Course No.	Course Title	No. of Credits	Credit Hours
MATH 1107	Calculus I	Theory: 3	15 × (1×3) = 45

Course Contents:

Calculus of single variable functions: Functions, Graphs and Algebra of Functions, Limit and continuity, Rules of Finding Limits; The Derivatives of Functions: Tangent line, Local Linear approximation and differential, Differentiation Rules, Chain Rule, Implicit Differentiation, Related Rates of Change, Mean Value Theorem, Partial derivatives, Taylor series with

Remainder, Indeterminate forms and L'Hopital's rule, Interval of increasing/decreasing, concavity, Extreme Values of Functions, First Derivative and Second Derivative Tests for Extreme Values, Optimization; Integrals: Indefinite Integrals, Integration by Parts, Integration by Substitution, Integration by Partial fractions, Riemann Sums, Definite Integral, Fundamental Theorem of Calculus, Average value of a function and Mean Value Theorem of integration, Substitution in Definite Integrals, Areas between Curves, Surface area and Volume of the surface of Revolution, Lengths of Plane Curves, Improper Integrals, Infinite series and convergence tests. Numerical Methods: Bisection, False Position, Newton-Raphson, Secant method, Fixed point iteration methods for approximating solutions of nonlinear equations, Newton's method for approximating solutions of systems of nonlinear equations.

References:

1. Howard Anton, Calculus, John Wiley & Sons, Inc.
2. G.B. Thomas and R.L. Finney, *Calculus and Analytical Geometry*, Addison Wesley, 9th Ed.
3. Erwin Kreyszig, *Advanced Engineering Mathematics*, John Wiley & Sons, 9th Ed.

Course No.	Course Title	No. of Credits	Credit Hours
GE 1105	Introduction to Sociology	Theory: 3	15 × (1×3) = 45

Course Contents:

Foundation of Sociology: Defining society and sociology. Sociology as a science. Origin and development of sociology as an academic discipline: Enlightenment, Industrial Revolution, French Revolution, and The United States Bill of Rights. **Theoretical Approaches to Sociology:** Major sociological perspectives: positivist, conflict, interactionist, critical, and postmodernist. Classical Theorists: August Comte, Karl Marx, Emile Durkheim, Max Weber and Du Bois. Contemporary Theorists: Antonio Gramsci, Herbert Marcuse, Charles W Mills, Pierre Bourdieu, Zygmunt Bauman, Manuel Castells, and Anthony Giddens. **Sociological Research Methods:** Defining scientific research in sociology. Research methodologies: qualitative (interview and ethnography) and quantitative (survey). Basic steps of scientific sociological research: defining a research problem, reviewing existing literature, developing research questions or formulating hypotheses, designing research, collecting data, analyzing findings (using SPSS for quantitative data and NVivo for qualitative data), and writing a report. **Society, Social Institutions and Social Organizations:** Evolution of modern human civilization. Types of society: ancient, feudal, and capitalist. Types of social institutions: marriage, family, education, and religion. Forms of social organizations: state, party, market, class, and status groups. **Social Stratifications and Social Problems:** Forms of social inequalities- slavery, estate, race, ethnicity, class, caste, gender, and sex. Social problems: war, crime, violence, ethnic conflict, drug trafficking, human trafficking, drug abuse, delinquency, and terrorism; displacement, dispossession, poverty, famine, malnutrition, disabilities, marginality, exclusion, and illiteracy. **Social and Environmental Changes:** Forms of globalization- economic, political, social and cultural. Modernization of agriculture, urbanization, industrialization, migration, and remittance. Digital civilization: fourth and fifth

industrial revolutions, information technology, and network society. Environmental crises: greenhouse effect, carbon emission, deforestation, drought, sea-level rise, salinity, flood, and cyclones; water and air pollution, soil erosion, and other environmental degradations.

References:

1. Sociology, 16th Edition, 2017 (Global Edition), by John J. Macionis.

Course No.	Course Title	No. of Credits	Credit Hours
SE 1106	Introduction to Software Engineering	Theory: 3	15 × (3×1) = 45

Course Contents: Introduction; Overview of Software Engineering: history, nature, relation of software engineering to other disciplines, software development life cycle, Programming language. Software nature and qualities: product qualities, project qualities, correctness, robustness, usability, maintainability, portability, quality measurements; Software development life cycle: requirement, design, development, testing, maintenance; Software development models: waterfall, agile, spiral, RDD, V model. Software measurement: modularity, abstraction, generality, object oriented, component oriented, structured. Specification and Verification: requirement specification, descriptive specification, testing, analysis, debugging; Modeling and Design: basics of modeling diagram, UI design. Software Project Management: concepts project metrics, estimation risks management.

References:

1. Software Engineering: A Practitioner's Approach, 7th Edition, McGraw Hill Higher Education.
2. Software Engineering, 7th edition Sommerville.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 1101L	Structured Programming Lab	1.5	15 × (1.5×2) = 45

Course Contents: Implementation and relevant lab work of the corresponding theory course.

References:

1. Software Engineering: A Practitioner's Approach, 7th Edition, McGraw Hill Higher Education.
2. Software Engineering, 7th edition, Sommerville.

Semester 2 (1st year 2nd Semester)

Course No.	Course Title	No. of Credits	Credit Hours
CSE 1215	Data Structure	Theory: 1.5	15 × (1.5×1) = 22.5

Course Contents: Data Structures and Complexity of Algorithms, Time Space Tradeoff, Searching Techniques: Linear and Binary Searching; Sorting and Recursion - Discussion of Common Sorting Techniques: Insertion Sort, Selection Sort, Bubble Sort, Quick Sort, Merge Sort, Radix Sort; Factorial and Tower of Hanoi Problem; Linked Lists - Abstract Data Types (ADT), List ADTs, and Linked Lists: Singly, Two Way and Circular Linked Lists; Stacks and Queues - Stacks and Queues and their Implementation Strategies;

References:

1. *Data Structures*. Schaum's Outline Series Effective Java, 2nd Edition, Joshua Bloch, Addison-Wesley
2. E. Horowitz and S. Sahni, *Fundamentals of Data Structures*, London Pitman
3. Robert L. Kruse, *Data Structures and Program Design*, Prentice Hall, 2nd Ed.

Course No.	Course Title	No. of Credits	Credit Hours
MATH 1216	Calculus II	Theory: 3	15 × (1×3) = 45

Course Contents:

Calculus of multivariable variable functions (mostly two variables). Functions, Graphs of Functions, Limit and continuity, Rules of Finding Limits, The Partial Derivatives of a Function, Tangent plane, Local Linear approximation and differential, Chain Rules, Related Rates of Change, Taylor series with Remainder, Extreme Values of Functions, First Derivative and Second Derivative Tests for Extreme Values, Optimization, Vectors in space, Dot and Cross products, Lines and Planes in Space, Lagrange multiplier technique for Constrained optimization. Gradient Descent method, Steepest Descent Method, Multiple Integrals with some applications. Differential Equations and Mathematical Modeling: First Order Initial Value Problems, Homogeneous Linear Equations with Constant Coefficients, Non-homogeneous Equations and Variation of Parameters. Critical Points, Stability of critical points, Qualitative Methods for Linear and Nonlinear Linear System of ordinary differential equations. Numerical Methods: Euler Method, Range-Kutta Method of order two and four for IVP.

References:

1. S.L. Ross, Differential Equation.
2. D.G.Zill, A First Course in Differential Equations with Applications.
3. Numerical Methods, E Balagurusamy, Tata McGraw-Hill Publishing Company.
4. Numerical Methods, Rao V. Dukkipati, New Age International Ltd.
5. Numerical Analysis, Richard L. Burden & J. Douglas Faires, Cengage Learning.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 1211	Computer Organization	Theory: 3	15 × (3×1) = 45

Course Contents: Function and structure of a computer, Functional components of a computer, Interconnection of components, Performance of a computer; Number representation and arithmetic: Binary, octal, and hexadecimal numbers, One's and two's complements and other representations, Addition and subtraction; Digital logic and integrated circuits: Boolean algebra and truth tables, Boolean functions (Gates, Functions, Simplification), Integrated circuits (Combinational circuits - adders, shifters, decoders, multiplexers and ROM's; Flip-flops; Sequential circuits - registers, counters and RAM); Representation of Instructions: Machine instructions, Operands, Addressing modes, Instruction formats, Instruction sets, Instruction set architectures - CISC and RISC architectures; Introduction to Assembly Language: Programming with Assembly language, The assembly process, Linking and loading, Register-level debugging, Processing Unit: Organization of a processor - Registers, ALU and Control unit, Data path in a CPU, Instruction cycle, Organization of a control unit - Operations of a control unit, Hardwired control unit, Micro-programmed control unit; Memory Subsystem: Semiconductor memories, Memory cells - SRAM and DRAM cells, Internal Organization of a memory chip, Organization of a memory unit, Error correction memories, Interleaved memories, Cache memory unit - Concept of cache memory, Mapping methods, Organization of a cache memory unit, Fetch and write mechanisms, Memory management unit - Concept of virtual memory, Address translation, Hardware support for memory management; Input/Output Subsystem: Access of I/O devices, I/O ports, I/O control mechanisms - Program controlled I/O, Interrupt controlled I/O, and DMA controlled I/O, I/O interfaces - Serial port, Parallel port, PCI bus, SCSI bus, USB bus, Firewall and InfiniBand, I/O peripherals - Input devices, Output devices, Secondary storage devices; Multiprocessing Systems: Shared memory multiprocessor, Message-passing multiprocessor, Hardware multithreading

References:

1. D. A. Patterson and J. L. Hennessy, "Computer Organization and Design - The Hardware/Software Interface", Morgan Kaufmann, 1998.
2. C. Hamacher, Z. Vranesic and S. Zaky, "Computer Organization", McGraw-Hill, 2002.
3. W. Stallings, "Computer Organization and Architecture - Designing for Performance", Prentice Hall of India, 2002.
4. J.P. Hayes, "Computer Architecture and Organization", McGraw-Hill, 1998.

Course No.	Course Title	No. of Credits	Credit Hours
STAT 1203	Probability and Statistics for Engineers II	Theory: 3	15 × (3×1) = 45

Course Contents:

Hypothesis Testing: Tests Concerning the Mean of a Normal Population, Testing the Equality of Means of Two Normal Populations, Hypothesis Tests Concerning the Variance of a Normal Population, Hypothesis Tests in Bernoulli Populations and Tests Concerning the Mean of a Poisson Distribution. **Regression and Correlation Analysis:** Least Squares Estimators of the Regression Parameters, Distribution of the Estimators, Statistical Inference about the Regression Parameters, Coefficient of Determination and Sample Correlation Coefficient, Analysis of Residuals, Transforming to Linearity, Weighted Least Squares, Polynomial Regression, Multiple Linear Regression, Logistic Regression Models for Binary Output Data and Correlation Analysis. **Analysis of Variance:** One-way Analysis of Variance, Two-Factor Analysis of Variance: Introduction and Parameter Estimation, Testing Hypotheses, and Two-way Analysis of Variance with Interaction Problems. **The goodness of Fit Tests and Categorical Data Analysis:** Goodness of Fit Tests when All Parameters are Specified, Goodness of Fit Tests when All Parameters are Unspecified, Tests of Independence in Contingency Tables, Tests of Independence in Contingency Tables Having Fixed Marginal Totals and Kolmogorov-Smirnov Goodness of Fit Test for Continuous Data. **Nonparametric Hypothesis Tests:** Sign Test, Signed Ranked Test, Two-Sample Problem, and Runs Tests for Randomness. **Simulation, bootstrap statistical methods, and permutation tests:** Random numbers, The bootstrap methods, permutation tests, Generating discrete and continuous random variables.

References:

1. Sheldon M. Ross, Introduction to Probability and Statistics for Engineers and Scientists, Elsevier/Academic Press, 4th Ed.
2. Douglas C. Montgomery and George C. Runger, Applied Statistics and Probability for Engineers, John Wiley and Son, 4th Ed.

Course No.	Course Title	No. of Credits	Credit Hours
GE 1212	Bangladesh Studies	Theory: 3	15 × (3×1) = 45

Course Contents: History and Society of Bengal under the British rule and Pakistan rule: The impact of British and Pakistan rules on the economy and education of the people. Language Movement of 1952, Events Leading to the Mass Upsurge of 1969, War of Independence and the Emergence of Bangladesh in 1971, **Government and Politics:** Three branches of government- executive, legislative and judiciary, Formation and role of major political parties in Bangladesh

and Constitutional development of Bangladesh, **Study of Geography and Resources of Bangladesh:** Location, Area, Boundary, Ecological Settings, River System, Climate, People and Resources of Bangladesh, **Social Structure of Bangladesh:** Rural society, Urban society, Family, household, and kinship, Women's role in society **Culture of Bangladesh:** Language, Literature, Art and Culture of Bangladesh, **Economy of Bangladesh:** Major Economic Sectors, potentials of various sectors and their prospective challenges; foreign aid and development, role of donor agencies, role of NGOs, **Achievements in different sectors of Bangladesh:** Economy, Culture, Sports, etc., **Socio-economic problems and prospects of Bangladesh:** poverty, health issues, natural disaster, social stratification and gender discrimination.

References:

1. Islam, S. (2003). Banglapedia. National Encyclopedia, Asiatic Society of Bangladesh, Dhaka
2. Kibria, S. A. (1999). *Bangladesh at the Crossroads*. University Press Ltd.
3. Riaz, A. (2016). *Bangladesh: A Political History since Independence*. London: IB Taurus.

Course No.	Course Title	No. of Credits	Credit Hours
SE 1206	Object Oriented Concepts	Theory: 3	15 × (3×1) = 45

Course Contents:

Introduction to Object Oriented Concepts: Procedural vs Object Oriented (OO) Programming, What is an Object – Object Data and Behavior, What is a Class – Attributes, Methods and Messages; **The Anatomy of a Class:** The Name, Comments, Attributes, Constructors, Accessors, Public Interface methods and Private implementation methods; **Constructors:** Default constructor, Using multiple constructors; **Object Oriented principles:** Encapsulation, Inheritance, and polymorphism; **Encapsulation:** Data Hiding, Interfaces and Implementations, **Inheritance:** Superclasses and Subclasses, Abstraction and Is-a Relationships, Reusing Objects, Generalization and Specialization, How Inheritance Weakens Encapsulation; Polymorphism, **Composition:** Abstraction and Has-a Relationships; How to think in terms of Objects – Interface vs Implementation, Abstract thinking when designing Interfaces and Giving the user minimal Interface possible; **Object Oriented Design Principle:** SOLID - Single Responsibility Principle, Open/Close Principle, Liskov Substitution Principle, Interface Segregation Principle and Dependency Inversion Principle.; **User Interface Handling:** User Interface Design, Layout, Item, Mouse Listener, Event Listener, etc. ; **Data Storage:** Store Data (Variable and Objects) in File and Database; Exception Handling; Object Serialization and Deserialization; **Collections and Generics:** Objects in various collections such as List, Map, Vector, Tree, Dictionary, etc.; **Object and Threading:** Thread, Multithreading, Forking, Concurrency, Sleep, etc.

References:

1. The Object-Oriented Thought Process (4th Edition) Matt Weisfeld, Addison-Wesley, 2013
2. Java: The Complete Reference, Ninth Edition Herbert Schildt, McGraw Hill
3. Java: How to Program, Eleventh Edition Paul Deitel, Harvey Deitel, Pearson, 2017
4. Effective Java, Third Edition, Joshua Bloch, Addison-Wesley

Course No.	Course Title	No. of Credits	Credit Hours
CSE 1215L	Data Structure Lab	Lab: 1.5	$15 \times (1.5 \times 2) = 45$

Course Contents: Review of data structures and algorithms: Solving problems (described scenarios) using data structures and algorithms in different languages such as C++, Java, Python, C# etc.; Implementing best searching techniques (for some specified scenarios): Linear and Binary Searching; Sorting and Recursion - Solving problems using common techniques: Insertion Sort, Selection Sort, Bubble Sort, Quick Sort, Merge Sort, Radix Sort; Implementing Factorial, Tower of Hanoi and similar type problems; Problem solving using data structures: Linked Lists - Abstract Data Types, List ADTs, and Linked Lists: Singly, Two Way and Circular Linked Lists; Stacks and Queues - Stacks and Queues and their Implementation Strategies.

References:

1. *Data Structures*. Schaum's Outline Series Effective Java, 2nd Edition, Joshua Bloch, Addison-Wesley
2. E. Horowitz and S. Sahni, *Fundamentals of Data Structures*, London Pitman
3. Robert L. Kruse, *Data Structures and Program Design*, Prentice Hall, 2nd Ed.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 1211L	Computer Organization Lab	Lab: 0.75	$15 \times (0.75 \times 2) = 22.5$

Course Contents: Implementation and relevant lab work of the corresponding theory course.

References:

1. D. A. Patterson and J. L. Hennessy, "Computer Organization and Design - The Hardware/Software Interface", Morgan Kaufmann, 1998.
2. C. Hamacher, Z. Vranesic and S. Zaky, "Computer Organization", McGraw-Hill, 2002.
3. W. Stallings, "Computer Organization and Architecture - Designing for Performance", Prentice Hall of India, 2002.
4. J.P. Hayes, "Computer Architecture and Organization", McGraw-Hill, 1998.

Course No.	Course Title	No. of Credits	Credit Hours
SE 1206L	Object Oriented Concepts Lab	Lab: 1.5	15 × (1.5×2) = 45

Course Contents:

The lab part will be designed based on any Object Oriented Programming Language. In each lab class, students will be given a scenario/ task where the realization of Object Oriented Concepts along with OO Language syntax will be implemented. Lab topics are given below.

Introduction and Environment Preparation: Introduction and installation of OO Language, Integrated Development Environment (IDE); Program compiling and running OO program, using classpath and Architecture; **Syntax** – Package, Import, Class, Fields, Methods, Constructors, Primitive data types, condition, iteration, Strings and literal, Nonexistence type: null. Class type: normal, inline, abstract, interface, etc. Programming features to support practical OOP – String Operations: String creations and operations, immutability property of String, String comparison and searching, String buffers and builders; Object and data I/O: Streams, Input and Output Stream, File, Path, Directory and tree; Exception handling: try and catch, checked exception vs unchecked exceptions, throw and throws, Common exception and User defined exceptions; Logger and Debugging: Logger, Log levels, Formatters and Filters, Logger Handlers and Manager; Generics: Wildcard, Generic class definitions, Generic method definitions, Using generics; Collection Framework: Collection interfaces, List and SortedList, Map and SortedMap, Navigable Map, Set and Sorted Set, Navigable Set, Queue and DeQueue, Stack; Data Storage: save in file, CRUD operation in any SQL database; Implementing User Interface; Multi-Threaded Programming: Overview of Thread, Creating and Running Thread, Thread Pools, Thread Synchronization, wait and notify, join and sleep and The concurrency API; Object Serialization and Deserialization.

References:

1. The Object-Oriented Thought Process (4th Edition) Matt Weisfeld, Addison-Wesley, 2013
2. Java: The Complete Reference, Ninth Edition Herbert Schildt, McGraw Hill
3. Java: How to Program, Eleventh Edition Paul Deitel, Harvey Deitel, Pearson, 2017
4. Effective Java, Third Edition, Joshua Bloch, Addison-Wesley.

Semester 3 (2nd year 1st Semester)

Course No.	Course Title	No. of Credits	Credit Hours
CSE 2115	Introduction to Algorithm	Theory: 3	15 × (1×3) = 45

Course Contents: Introduction - Algorithms, Analyzing & Designing Algorithms, Correctness of Algorithms; Greedy Algorithms - Introduction to Greedy Algorithms, Greedy Choice Property, Greedy vs. Dynamic Programming, Fractional Knapsack Problem, Activity Selection Problem, Huffman Encoding, Task Scheduling Problem, Coin Changing Problem, Kruskal's and Prim's Minimum Spanning Tree Algorithms; Divide and Conquer Algorithms - Introduction to Divide and Conquer Design Technique, Quick Sort, Merge Sort, Proof of Correctness, and Run Time Analysis; Dynamic Programming - Introduction to Dynamic Programming Technique, Principle of Optimality, Optimal Substructure Property, Assembly Line Scheduling, Matrix Chain Multiplication, LCS, Viterbi Algorithm, Bitonic Euclidean Traveling Salesperson Problem and Runtime Analysis; Graph Searching and Shortest Path Problems - Breadth First Search, Depth First Search, Flow Networks, Single Source and All Pair Shortest Path Algorithms; Selected Topics - String Matching Algorithms; NP Completeness and Approximation Algorithms - NP Completeness, Polynomial Time Verification, NP Completeness and Reducibility, NP Complete Problems and Approximation Algorithms.

References:

1. Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein.

Course No.	Course Title	No. of Credits	Credit Hours
SE 2112	Theory of Computing	Theory: 3	15 × (1×3) = 45

Course Contents:

Basic concepts and definitions; Set operations; partition of a set; Equivalence relations; Properties on relation on set; Proving Equivalences about Sets; Central concepts of Automata Theory; Deterministic Finite Automata (DFA); Minimization of DFA; Non-Deterministic Finite Automata (NFA); Equivalence of Deterministic and Non-Deterministic Finite Automata; Equivalence between DFA, NFA, NFA; Regular Expressions; Operations on Regular expressions Finite Automata and Regular Expressions; Conversion from FA and regular expressions; Context-Free Grammars; Parse Trees; Ambiguity in Grammars and Languages; Standard Forms; Chomsky Normal Forms; Greibach normal Forms; Minimization of CFG's; Pushdown Automata (PDA); Deterministic and Non-Deterministic (PDA); Formal definition of

NPDA; Transition functions of NPDA; NPDA Execution; Accepting Strings with NPDA; Equivalence of PDAs and CFG; The Turing Machine.

References:

1. John Hopcroft, Rajeev Motowani, and Jeffrey Ullman, *Automata Theory, Languages, and Computation*. (3rd Edition)
2. Introduction to Languages and the Theory of Computation - 4th edition John Martin, 2006.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 2111	Computer Networking	Theory: 3	15 × 3 = 45

Course Contents: Overview of the Internet, Overview of Networking Protocols, Network Edge, Network Core, Protocol Layers, Internet Structure, Delays; Application Layer: Principles of Networking Applications, Web and HTTP, FTP, SMTP, DNS; Transport Layer: Transport Layer Services, Multiplexing and De multiplexing, Connection-less Transport: UDP, Connection-Oriented Transport: TCP, Principles of Congestion Control, TCP Congestion Control; Network Layer: Inside a Router, Details of the Internet Protocol (IP), IPv4 Subnetting, SDN Concepts, Routing Algorithms (Link State, Distance Vector), Intra-AS routing in the Internet: Open Shortest Path First (OSPF), Routing among the ISPs: Border Gateway Protocol (BGP); Link Layer: Multiple Access Protocols, Switched LAN, 802.11 Wireless LANs

References:

1. Computer Networking- A top down Approach -Kurose and Ross
2. Data Communication & Networking – Behrouz Forouzan- McGraw Hill Education
3. Computer Network –Tannenbaum – Pearson Education

Course No.	Course Title	No. of Credits	Credit Hours
MATH 2116	Linear Algebra	Theory: 3	15 × (1×3) = 45

Course Contents:

Matrix Algebra, Determinant and its properties, Inverse of a square matrices and properties. Elementary matrices and reduction of a matrix to row echelon form and reduced row echelon forms and their application in solving system of linear equations. Vector space and subspaces of a vector space, Linear span and independence of vectors, basis and dimension of a vector space. Row space, Colum Space, Null space, and Null space of the transpose of a matrix. Orthogonality, Orthogonal projection matrices, Orthonormal bases, Gram-Schmidt orthogonalization algorithm and QR factorization. Eigenvalues, Eigenvectors, Orthonormal diagonalization, Diagonalization of symmetric matrices. Normal and Positive definite matrices. Linear transformation and their matrices, Change of bases, Geometrically important linear

transformations. Singular Value Decomposition (SVD), Geometric interpretation of SVD, Principal Component Analysis (PCA) from SVD. Jacobi and Gauss-Seidel methods for approximating solutions of linear system. Power Method for approximating Eigenvalues and Eigenvectors.

References:

1. Linear Algebra with Applications, 8th Edition, Steven J. Leon.
2. Introduction to Linear Algebra, 5th Edition, Gilbert Strang.

Course No.	Course Title	No. of Credits	Credit Hours
SE 2105	Software Project Lab I	Lab: 3	$15 \times (3 \times 2) = 90$

Introduction of the Course: Each of the students should complete the software project individually. Students are required to develop sufficiently large software which requires significant “problem solving” effort. Besides, students must showcase the skills they have acquired from their so far completed courses.

Specific Objectives:

- To master coding skills
- To learn to create a complete software
- To learn to manage large codebase
- To keep track of changes to the code through version controlling systems

Prerequisite: SE 1206, CSE 1215, STAT 1203, MATH 1216

Course No.	Course Title	No. of Credits	Credit Hours
BUS 2117	Business Communication	Theory: 3	$15 \times 3 = 45$

Course Contents:

Communication Concept: The Role of Communication in Business, Importance of Communication Skills, Main Form of Business Communication, Process of Human Communication. **Fundamentals of Business Writing:** Adaptation and the Selection of Words, Construction of Clear Sentences and Paragraphs, Writing for Effect. **Basic Pattern of Business Messages:** Directness in Good News and Neutral Situations, Indirectness in Bad Message, Indirectness in Persuasion Message, Letter and Memorandum, Letter Writing Styles, Pattern Variations in Memorandums and the Email, Job Search Activities: Strategies in the Job Search Process, Job search activities, Writing CV, Facing Interviews, Feedback letters for goodwill, **Fundamentals of Report Writing:** Basics of Report Writing, Report Structure: Short Forms, Long and Formal Report, Usages of Graphics. **Other Form of Business Communication:** Informal Oral Communication, Technology-Enabled Communication. **Psychology in Business:**

Job Analysis: Job-oriented Approach, Person-oriented Approach, Purposes of Job Analysis, Methods of Job Analysis, Job Evaluation, Assessment Methods for Selection and Placement: Productive and Counterproductive Employee Behavior: Productive Behavior, Job Performance; Counterproductive Behavior, Withdrawal, Aggression, Mistreatment, Sabotage, and Theft.

References:

1. Industrial and Organizational Psychology: Research and Practice, Paul E. Spector, 5th Edition
2. Raymond V. Lesikar, John D. Pettit, Maire E. Flatley, Lesikar's Basic Business Communication, McGraw Hill

Course No.	Course Title	No. of Credits	Credit Hours
CSE 2115L	Introduction to Algorithm Lab	Lab: 1.5	$15 \times (1.5 \times 2) = 45$

Course Contents: Review of Algorithms, Analyzing & Designing Algorithms, Greedy Algorithms - Introduction to Greedy Algorithms, Greedy Choice Property, Dynamic Programming, Fractional Knapsack Problem, Activity Selection Problem, Huffman Encoding, Task Scheduling Problem, Coin Changing Problem, Kruskal's and Prim's Minimum Spanning Tree Algorithms; Divide and Conquer Algorithms - Introduction to Divide and Conquer Design Technique, RunTime Analysis; Dynamic Programming - Introduction to Dynamic Programming Technique, Principle of Optimality, Optimal Substructure Property, Assembly Line Scheduling, Matrix Chain Multiplication, LCS, Viterbi Algorithm, Bitonic Euclidean, Traveling Salesperson Problem; Graph Searching and Shortest Path Problems - Breadth First Search, Depth First Search, Flow Networks, Single Source and All Pair Shortest Path Algorithms; String Matching Algorithms, Approximation Algorithms.

References:

1. Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 2111L	Computer Networking Lab	Lab: .75	$15 \times (.75 \times 2) = 22.5$

Course Contents: Implementation and relevant lab work of the corresponding theory course.

References:

1. Computer Networking - A top down Approach, Kurose and Ross.
2. Data Communication & Networking, Behrouz Forouzan, McGraw Hill Education.
3. Computer Network, Tannenbaum, Pearson Education.

Semester 4 (2nd year 2nd Semester)

Course No.	Course Title	No. of Credits	Credit Hours
CSE 2201	Operating System and System Programming	Theory: 3	15 × (1×3) = 45

Course Contents:

Introduction: What is an operating system? History of operating system Operating system concepts Operating system structure; **Processes and Threads:** Processes state, operations on process, inter-process communications Threads and concurrency Process and thread scheduling **Process synchronization:** Synchronizations tools, Principles of Concurrency, Critical Region, Mutual Exclusion, Semaphores and Mutex, Monitors, Classical Problems of Synchronization: Readers-Writers Problem, Producer Consumer problem, dining philosopher problem. **Deadlock:** Principles of deadlock, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection Recovery from deadlock, An Integrated Deadlock Strategies, Other Issues: Two phase locking, Communication Deadlock, Livelock, Starvation; **Memory Management:** Memory address, Swapping and Managing, Free Memory Space Virtual Memory Management, Paging Segmentation, Paged Segmentation, Demand Paging Performance, Page Replacement Algorithms, Allocation of Frames, Thrashing; **File Systems:** Files, Directories, File system management, Input /Output, Principles of I/O, hardware Principles of I/O, software I/O, software layers, Disks.

References:

1. Silberschatz A., Galvin P., Gagne G., "Operating System Concepts", 10th Edition, John Wiley and Sons,
2. Andrew S. Tanenbaum, "Modern Operating Systems", 5th Edition, PHI
3. Stallings William, "Operating Systems", 8th Edition, Pearson Education
4. Milan Milenkovic, "Operating Systems Concepts and Design", TMGH
5. Das Sumitabha, "Unix Concepts and Applications", 3rd Edition, Tata McGraw Hill, 2003

Course No.	Course Title	No. of Credits	Credit Hours
SE 2215	Design Patterns	Theory: 1.5	15 × (1.5×1) = 22.5

Course Contents:

Concepts of Object Oriented Design and Design Principles: Revision of OO Concepts, Using UML Diagrams, aggregation, composition and association, Coupling and Cohesion,

(SOLID) - Single responsibility principle, Open-Closed Principle, Liskov substitution principle, Interface segregation principle, Dependency Inversion principle. **Refactoring code smells:** Different type of code smells - Inappropriate Naming, Comments, Dead Code, Duplicated code, Primitive Obsession, Large Class, Lazy Class, Alternative Class with Different Interface, Long Method, Long Parameter List, Switch Statements, Speculative Generality, Oddball Solution, Feature Envy, Refused Bequest, Black Sheep and Train Wreck, **Design Patterns:** Importance of learning design patterns, Types of Design Patterns - Structural, Behavioral and Creational Patterns, Creational Patterns – Singleton, Factory, Factory Method, Abstract Factory, Builder, Prototype and Object Pool, Behavioral Patterns - Chain of Responsibility, Command, Interpreter, Iterator, Mediator, Memento, Observer, Strategy, Template Method, Visitor and Null Object, Structural Patterns – Adapter, Bridge, Composite, Decorator, Flyweight and Proxy.

References:

1. Gamma, Erich. *Design patterns: elements of reusable object-oriented software*. Pearson Education, 1995.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 2204	Database Management System	Theory: 3	15 × (1×3) = 45

Course Contents:

Introduction to Database Systems: Evolution of file processing systems, role of databases in organizations, core components of a database environment; Data Modeling: the Entity-Relationship Diagram; The Relational Model and Normalization: relational model, normalization, transformation of an entity-relationship data diagram into a relational model, express query in relational algebra; Indexing and Hashing: Indexing, B+ tree, static and dynamic hashing; Query Processing and Query Optimization: Measures of Query Cost, Selection Operation, Sorting, Join Operation, Statistical Information for Cost Estimation, Cost-based optimization; Transaction; Concurrency Control: Lock-Based Protocols, Timestamp-Based Protocols; Recovery System: Recovery and Atomicity, Log-Based Recovery; Database System Architectures: Centralized and Client-Server Systems, Server System Architectures; Parallel Databases:I/O Parallelism, Interquery Parallelism. Intraquery Parallelism, Intraoperation Parallelism, Interoperation Parallelism; Distributed Databases: Distributed Data Storage, Distributed Transactions, Commit Protocols, Concurrency Control in Distributed Databases.

References:

1. Avi Silberschatz, Henry F. Korth, S. Sudarshan; Database System Concepts, 5th Edition
2. Raghu Ramakrishnan, Johannes Gehrke' Database Management Systems, 3rd Edition

Course No.	Course Title	No. of Credits	Credit Hours
BUS 2205	Business Studies for Engineers	Theory: 3	15 × (3×1) = 45

Course Contents: Managers and Entrepreneurs: Management Defined, Role of a Manager, Small-Business Management, The Evolution of Management Thought, Organization, Organization Charts, Contrasting Theories of Organization, Organizational Effectiveness, Organizational Cultures, Change, Conflict, and Negotiation in Organization; The Strategic Management Process, Strategic Implementation and Control, Forecasting. **Accounting Basic:** Forms of Business Organization, Types of Activities performed by Business Organization, Financial statements of Business Organization, The Accounting Equation, The Account and Rules of Debit and Credit, The Journal: Recording of Transaction, Adjusting the Accounts, Closing Entries, and Preparing Financial statements from the Work Sheet. **Analysis and Interpretation of Financial Statement:** Objectives of Financial Statement Analysis, Analysis of a Balance Sheet, Analysis of Statement of Income and Retained Earnings, Ratio Analysis: Liquidity Ratios, Equity or Long Term Solvency Ratio, Profitability Test, Market Test.

References:

1. Stephen P. Robbins and Mary Coulter, *Management*, Prentice Hall, Latest Edition
2. Jerry J. Weygandt, Donald E. Kieso, and Paul D. Kimmel, *Accounting Principles*, Wiley, 8th Ed

Course No.	Course Title	No. of Credits	Credit Hours
SE 2206	Software Requirements Specification and Analysis	Theory: 1.5	15 × (1×1.5) = 22.5

Course Contents: Review of – The Nature of Software, Software Engineering, The Software Process, Software Engineering Practices, Generic Software Process Model, Process Assessment and Improvement, Prescriptive Process Models, Specialized Process Model and Agile Development. Requirements Engineering, Establishing the ground work, Eliciting Requirements, Negotiating Requirements, Validating Requirements, Requirements Analysis, Scenario-Based Modeling, UML Models, Class Based Modeling, Requirements Modeling Strategies, Flow-Oriented Model, Behavioral Model.

References:

1. R. S. Pressman, *Software Engineering. A Practitioner's Approach*, 7th or higher, McGraw Hill
2. Ian Sommerville. *Software Engineering*, 9th or higher Edition, Addison-Wesley.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 2201L	Operating System and System Programming Lab	Lab: 1.5	15 × (1.5×2) = 45

Course Contents:

Basic of Unix Shell commands, Programs using the system calls of UNIX operating system fork, exec, getpid, exit, wait, close, stat, opendir, readdir; Introduction to shell programs; Process creation and thread creation. Problem solving using process and thread; Implementation of CPU scheduling algorithms, FCFS, SJF, Priority and Round Robin algorithm; Introduction to MUTEX, semaphore and implementation of classical synchronization problems using semaphore; Inter Process Communication using shared memory; Implementation of Banker's algorithm for deadlock detection; Implementation of memory allocation methods using fixed partitioned; Implementation of paging techniques; File & Buffered I/O, directory management in Linux, Basic Bash scripting in Linux, Socket programming in C; Writing a simple device driver.

References:

1. Linux System Programming, Talking directly from the Kernel and C Library- by Robert Love (2nd edition)

Course No.	Course Title	No. of Credits	Credit Hours
SE 2215L	Design Patterns Lab	Lab: 1.5	15 × (1.5×2) = 45

Course Contents:

Implementing a scenario following the SOLID principle; Refactoring the following type of code smells from existing open source projects: Inappropriate Naming, Comments, Dead Code, Duplicated code, Primitive Obsession, Large Class, Lazy Class, Alternative Class with Different Interface, Long Method, Long Parameter List, Switch Statements, Speculative Generality, Oddball Solution, Feature Envy, Refused Bequest, Black Sheep and Train Wreck, Implementation of a scenario for the following design patterns: Creational Patterns – Singleton, Factory, Factory Method, Abstract Factory, Builder, Prototype and Object Pool, Behavioral Patterns - Chain of Responsibility, Command, Interpreter, Iterator, Mediator, Memento, Observer, Strategy, Template Method, Visitor and Null Object, Structural Patterns – Adapter, Bridge, Composite, Decorator, Flyweight and Proxy.

References:

1. Gamma, Erich. Design patterns: elements of reusable object-oriented software. Pearson Education, 1995.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 2204L	Database Management System Lab	Lab: 1.5	$15 \times (1.5 \times 2) = 45$

Course Contents:

SQL: DDL and DML commands; query using select and where clause, rename operation, string operation, set operation, aggregate function, group by, having clause, nested query, in, all, some construct; PL/SQL: named and anonymous function, procedure, trigger, cursor; database administration: Creating Tables, Clusters and Database Tuning; Control File Maintenance; Managing Indexes; Privileges and Roles; SQL: Creation, modification, configuration, and deletion of databases using SQL, Creation of database schema - DDL (create tables, set constraints, enforce relationships, create indices, delete and modify tables); Database initialization - Data insert, Data import to a database (bulk import using SQL Commands), Practice SQL commands for DML (using select and where clause, rename operation, string operation, set operation, aggregate function, group by, having clause, nested query, in, all, some construct); PL-SQL: use of conditional statement, different types of Loops; use of function, procedure and cursor; database administration: Creating Tables, Clusters and Database Tuning; Control File Maintenance; Managing Indexes; Privileges and Roles;

References:

1. Avi Silberschatz, Henry F. Korth, S. Sudarshan; Database System Concepts, 5th Edition.
2. Raghu Ramakrishnan, Johannes Gehrke' Database Management Systems, 3rd Edition.

Course No.	Course Title	No. of Credits	Credit Hours
SE 2206L	Software Requirements Specification and Analysis Lab	Lab: 1.5	$15 \times (2 \times 1.5) = 45$

Course Contents:

One small to medium sized real life system will be given to all the students for analyzing in the classroom. Every week students have to bring their own models (Scenario, Data, Object and Behavioral) in the class and all the students will analyze and learn from each other's models.

Two real life mid-scale systems will be distributed among groups (created randomly) of 5/6 students to analyze (one project per group). The output of the analysis will be specification reports.

References:

1. R. S. Pressman, Software Engineering. A Practitioner's Approach, 7th Edition or higher, McGraw Hill.
2. Ian Sommerville. Software Engineering, 9th or higher Edition, Addison-Wesley

Semester 5 (3rd year 1st Semester)

Course No.	Course Title	No. of Credits	Credit Hours
SE 3115	Software Testing & Quality Assurance	Theory: 1.5	15 × (1.5×1) = 22.5

Course Contents:

The Psychology and Economics of Software Testing, Software Testing Life Cycle (STLC), Software Testing Terminology and Methodology, V&V Model, Dynamic Black Box Testing – Boundary Value Analysis, Equivalence Partitioning, State Transition based Testing, Decision Table based Testing, Cause-Effect Graphing based Testing and Error Guessing, Dynamic White Box Testing – Basis Path Testing, Data Flow Testing and Mutation Testing, Inspections, Walkthroughs, Technical Reviews, Unit Testing, Integration Testing, Function Testing, System Testing, Acceptance Testing, Regression Testing, Test Management – Test Organization, Test Plan, Test Design and Specifications, Software Metrics, Software Quality, Quality Control and Quality Assurance, Quality Management and Project Management, Software Quality Metrics, Testing Internet Applications - Security and Performance Testing, Debugging, Test Driven Development (TDD), Behavior Driven Development (BDD).

References:

1. Naresh Chauhan, Software Testing: Principles and Practices, 1st or higher Edition, Oxford University Press.
2. Glenford J. Myers, Corey Sandler, and Tom Badgett. The Art of Software Testing, 3rd or higher Edition, John Wiley & Sons.
3. Lisa Crispin and Janet Gregory. Agile Testing: A Practical Guide for Testers and Agile Teams, 1st or higher Edition, Pearson Education.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 3102	Web Technology	Theory: 1.5	15 × (1.5×1) = 22.5

Course Contents:

Introduction to Html, Java Script & CSS, Server Side Programming: HTTP Server, Application Server, MVC Web Framework, Web Services, Database Access: Object Relational Mapping,

Lambda Expression, Language Integrated Query, Data Reader, Writer, Web Security: Denial of Service, Buffer Overflow, Cross Site Scripting, Authentication and Access Control.

References:

1. Deitel & Deitel, Goldberg, “Internet and world wide web – How to Program”, Pearson Education Asia, 2001.
2. Rajkamal, “Web Technology”, Tata McGraw-Hill, 2001.
3. Teach yourself web technologies part I & II- I. Bayross. BPB
4. Web Design in a Nutshell- J. Niederst, SPD

Course No.	Course Title	No. of Credits	Credit Hours
CSE 3116	Mathematical and Statistical Methods for Data Analysis	Theory: 3	$15 \times (3 \times 1) = 45$

Course Contents:

Probability and distribution: . Hypothesis Testing: ANOVA, . Stochastic Process: Markov chains, HMM. Matrix Factorization and Applications: QR factorization, LU decomposition, Singular value decomposition (SVD), Dimension reduction via PCA, SVD, graph partitioning, introduction to spectral graph theory, k-means clustering and its relation to matrix factorization. Advanced Regression Analysis: Linear least Squares, . Optimization: Normal equations and gradient descent. Information Theory: mutual information, entropy.

References:

1. Matrix Computations, 3rd Edition, Gene H. Golub, Charles F. Van Loan.
2. Linear Algebra with Applications, 8th Edition, Steven J. Leon.
3. Introduction to Linear Algebra, 5th Edition, Gilbert Strang.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 3117	Artificial Intelligence	Theory: 3	$15 \times (3 \times 1) = 45$

Course Contents:

The Foundations and history of Artificial Intelligence, The State of the Art Agents and Environments, The Concept of Rationality, Problem-Solving Agents, Uninformed Search, Informed (Heuristic) Search, State-of-the-Art Game Programs, Constraint Satisfaction Problems, Bayes' Rule and Its Use, Probabilistic Reasoning, Bayesian Networks, Inference in

Temporal Models, Hidden Markov Models, Dynamic Bayesian Networks, Propositional Logic, First-Order Logic, Knowledge Representation, Ontological Engineering, Supervised and unsupervised Learning, Learning Decision Trees, Artificial Neural Networks, Support Vector Machines, Ensemble Learning, Brief introduction to Language Models, Text Classification, Information Retrieval, Information Extraction, Syntactic Analysis (Parsing), Machine Translation, Speech Recognition, Perception, Robotic Perception, Moving.

References:

1. Russell, S., & Norvig, P. Artificial intelligence: a modern approach. Third Edition. Pearson new international edition. 2014.
2. Cawsey, A. (1998). The essence of artificial intelligence. Prentice Hall.
3. Poole, D. L. & Mackworth, A. K. (2010). Artificial intelligence: foundations of computational agents. Cambridge University Press.

Course No.	Course Title	No. of Credits	Credit Hours
SE 3118	Software Design and Analysis	Theory: 1.5	15 × (1.5×1) = 22.5

Course Contents: Theory:

Design Concept - The Design Process, Design Concepts, The Design Model; Architectural Design: Software Architecture, Architectural Genres, Architectural Styles, Architectural Design Process, Assessing Alternative Architectural Designs, Architecture and Agility; Component-Level Design: Definition of a Component, Designing Class-Based Components, Conducting Component-Level Design, Component-Level Design for WebApps, Designing Traditional Components, Component-Based Development; User Interface (UI)/ User Experience (UX) Design: The Golden Rules, Nielsen's Heuristics, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, Web App Interface Design, Design Evaluation.

References:

1. Software Engineering: A Practitioner's Approach, 8th Edition, Roger S. Pressman.
2. Software Engineering, 10th Edition, Ian Sommerville.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 3119	Cryptography and Security Mechanism	Theory: 3	15 × (1×3) = 45

Course Contents:

Introduction to Information Security: Security attacks [active, passive, brute force, cryptanalysis, insider], security services [CIA, AAA], security terminologies [adversary, vulnerability, threat, attack], assumption and trust, security policy and mechanism, threat analysis, attacker modeling, **Introduction to Cryptography:** Plaintext, ciphertext, encryption, decryption, symmetric and asymmetric encryptions, block cipher and stream ciphers, historical cipher techniques. **Modern Cryptography:** DES, need for 3DES, computationally secure cipher, unconditionally secure cipher, types of attack on cryptographic text, cipher mode of operation, Cryptographic algorithm: Detail operation of AES, RC4 Algorithm, **Cryptographic algorithm:** Introduction to modular arithmetic, detail operation of RSA, correctness proof of RSA, **Hash:** Properties of Hash functions, hash algorithm variants, SHA 512, Introduction to HMAC and CMAC, Authentication Mechanism, Password-based authentication, Biometric authentication, Challenge-Response authentication, Multifactor authentication, Access Control and Different Access Control Models, Diffie Hellman (DH) key exchange protocol, Problems in DH key exchange protocol, Public-Key Infrastructure and Digital Signature (Generation and Verification), Risk Management Framework.

References:

1. Cryptography and Network Security- Principles and Practice, William Stallings
2. Computer Security- Principles and Practice, William Stallings and Lawrie Brown
3. Introduction to Computer Security, Matt Bishop
4. Security in Computing, Charles P. Pfleeger, Shari Lawrence Pfleeger and Jonathan Margulies
5. Computer Security: Art and Science, Matt Bishop
6. Network Security: Private Communication in a Public World, Charlie Kaufman, Radia Periman, and Mike Speciner.

Course No.	Course Title	No. of Credits	Credit Hours
SE 3105	Software Project Lab II	Lab: 3	15 × (2×3) = 90

Introduction of the Course: Software engineering ensures quality by following rigorous processes throughout the Software Development Life Cycle (SDLC). Requirement engineering, or the engineered process of collecting, analyzing, specifying and validating requirements is considered the most crucial of all these processes as it determines the functionalities of the software. The goal of SE 3105: Software Project Lab II (SPL II) is to ensure that students follow and apply the processes of requirement engineering while creating software. Therefore, it is necessary that students choose and propose a project that will allow them to apply requirement engineering within the time frame of the project. Moreover, it is also encouraged and suggested that students choose projects with ample scope to gather requirements from stakeholders outside their comfort zone, i.e. from IIT, relatives, and/or friends. Students are

required to create a team of 2 members. However, the number of team members can be varied for special cases which will be decided by the assigned course manager.

Specific Objectives:

- To acquire the ability to collect requirements for real-world projects
- To learn to work collaboratively
- To be able to manipulate data intensive projects

Prerequisite: SE 2206, SE 2215.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 3102L	Web Technology Lab	Lab: 1.5	$15 \times (1.5 \times 2) = 45$

Course Contents:

HTML: Basic tags of HTML, Forms and Controls, CSS: Different types of CSS tags and their values, Java Script: Elements of Java Script, different types of objects in JS, DOM manipulation, event handling, Server Side Programming: Introduction to ASP.NET and Visual Studio IDE, Web development using ASP.NET MVC, Database Programming using Entity Framework with Lambda Expression and Language Integrated Query (LINQ), Web Services.

References:

1. Deitel & Deitel, Goldberg, “Internet and world wide web – How to Program”, Pearson Education Asia, 2001.
2. Rajkamal, “Web Technology”, Tata McGraw-Hill, 2001.
3. Teach yourself web technologies part I & II- I. Bayross. BPB
4. Web Design in a Nutshell- J. Niederst, SPD

Course No.	Course Title	No. of Credits	Credit Hours
CSE 3119L	Cryptography and Security Mechanisms Lab	Lab: 0.75	$15 \times (0.75 \times 2) = 22.5$

Course Contents: Implementation and relevant lab work of the corresponding theory course.

References:

1. Cryptography and Network Security- Principles and Practice, 5th Edition, William Stallings
2. Computer Security- Principles and Practice, 2nd Edition, William Stallings and Lawrie Brown
3. Security in Computing - 2nd Edition, Charles P. Pfleeger and Shari Lawrence Pfleeger
4. Introduction to Computer Security- 1st Edition, Matt Bishop
5. Computer Security: Art and Science- 1st Matt Bishop
6. Network Security: Private Communication in a Public World- 2nd Edition- Charlie Kaufman, Radia Periman, and Mike Speciner.
7. Computer Networking- A top down Approach (6th Edition) -Kurose and Ross

Semester 6 (3rd year 2nd Semester)

Course No.	Course Title	No. of Credits	Credit Hours
SE 3215	Internship	Lab: 15	15 × (2×15) = 450

Introduction of the Course: To gain practical experience, students of the sixth semester will be assigned to a professional establishment as interns. Students, upon completion of 5th semester, will have an acceptable level of theoretical knowledge, which they will be able to apply in practice. At the same time, they will become aware of the practical trends and professional environment. After completion of the internship, students will have to present a report on the respective domain of work.

Specific Objectives:

- To learn to work in a professional environment
- To improve the communication, team-work and other relevant soft skills
- To apply theoretical knowledge to real-life projects

Semester 7 (4th year 1st Semester)

Course No.	Course Title	No. of Credits	Credit Hours
CSE 4115	Distributed Systems	Theory: 1.5	$15 \times (1.5 \times 1) = 22.5$

Course Contents:

Foundations: Characterization of DS, System Models, Networking and Inter-networking, Inter-process Communication, Remote Invocation, Indirect Communication and Operating System Support; Middle-ware: Distributed Objects and Components, Web Services and Peer-to-Peer Systems services, Security, Distributed File Systems and Name Services; Distributed algorithms: Time and Global States, Coordination and Agreement Shared data, Transactions and Concurrency Control, Distributed Transactions, and Replication, New challenges: Mobile and Ubiquitous Computing.

References:

1. Distributed Systems: Principles and Paradigms , Tanenbum.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 4116	Compiler Design	Theory: 3	$15 \times (1 \times 3) = 45$

Course Contents:

Introduction to Compilers: Structure of a compiler – Lexical Analysis – Role of Lexical Analyzer – Input Buffering –Specification of Tokens – Recognition of Tokens – Lex – Finite Automata – Regular Expressions to Automata – Minimizing DFA. **Syntax Analysis:** Role of Parser – Grammars – Error Handling – Context-free grammars – Writing a grammar –Top Down Parsing – General Strategies Recursive Descent Parser Predictive Parser-LL(1) Parser-Shift Reduce Parser-LR Parser-LR (0)Item Construction of SLR Parsing Table -Introduction to LALR Parser – Error Handling and Recovery in Syntax Analyzer-YACC. **Intermediate Code Generation:** Syntax Directed Definitions, Evaluation Orders for Syntax Directed Definitions, Intermediate Languages: Syntax Tree, Three Address Code, Types and

Declarations, Translation of Expressions, Type Checking. **Run-Time Environment and Code Generation:** Storage Organization, Stack Allocation Space, Access to Non-local Data on the Stack, Heap Management – Issues in Code Generation – Design of a simple Code Generator.

Code Optimization: Principal Sources of Optimization – Peep-hole optimization – DAG-Optimization of Basic Blocks, Global Data Flow Analysis – Efficient Data Flow Algorithm.

References:

1. Compilers: Principles, Techniques and Tools, Second Edition, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffry D. Ullman.
2. Lex & Yacc – John R. Levine, Tony Mason, Doug Brown, O'reilly
3. Compiler Construction, Loudon, Thomson.

Course No.	Course Title	No. of Credits	Credit Hours
SE 4117	IT Laws & Ethics	Theory: 3	15 × (3×1) = 45

Course Contents:

Introduction: Introduction to legal aspects, Jurisdiction, Intellectual Property Rights (Copyright and infringement, fair use Doctrine, Software copyright protection, Patent, software patent, software cross-licensing agreements, and Trademark), Contracts and licenses, E-contract, Privacy in the workplace, Trade secrets and non-disclosure agreement; **Computer and Internet Crime:** Threats to information resources, including military and economic espionage, Malware, **IT security incidents:** communications eavesdropping, computer break-ins, denial-of-service, destruction and modification of data, distortion and fabrication of information, forgery, etc. **Relevant Laws:** Cyber laws and rights in today's digital age, Relevant laws and policy guidelines (ICT Act 2006, Digital Security Act 2018, Cyber security strategy guide, Information Security policy for Bank and non-bank financial institutes), Laws relevant to Intellectual Property Rights and E-contract. **Introduction to Ethics:** Definition of Ethics, Moral code, morality, importance of integrity, vices, virtues, Difference between morals, ethics, and law, Fostering good business ethics, establishing corporate code of ethics, Knowledge about ethical work environment. **Ethics for IT Workers and IT Users:** IT Profession, Professional Relationships that IT workers must manage, Ethical issues aroused in IT organizations, Professional code of ethics, professional organizations, certifications, government licensing, issues associated with the government licensing IT Workers, IT Professional malpractices, Common Ethical issues for IT Users, supporting the ethical practice of IT Users. **Freedom of Expression:** Freedom expression, obscene speech, defamation, hate speech, key issues related with freedom of expressions, internet filtering, and anonymous communication. **Ethics in**

software development and for IT organizations: Social networking, Key ethical issues in social networking, digital divide, how to deal with a particular case of any ethical issues in a company. **E-Waste management.**

References:

1. George W. Reynolds, Ethics in Information Technology.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 4118	Machine Learning	Theory: 3	$15 \times (1 \times 3) = 45$

Course Contents:

Basic definitions, types of learning, evaluation, cross-validation, Linear and logistic Regression, Support Vector Machine, Decision trees and ensemble learning, Instance based learning, feature reduction, probability and bayes learning, Bayesian networks, Dimensionality Reduction, Clustering: k-means, adaptive hierarchical clustering, gaussian mixture model, Neural Networks: Perceptron, multilayer network, backpropagation, Introduction to deep neural networks.

Reference Books:

1. The Elements of Statistical Learning: Data Mining, Inference, and Prediction by Trevor Hastie, Robert Tibshirani, and Jerome Friedman.
2. Pattern Recognition and Machine Learning by Christopher M. Bishop.
3. Data Mining: Practical Machine Learning Tools and Techniques by Ian H. Witten, Eibe Frank, and Mark A. Hall.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 4115L	Distributed Systems Lab	Lab: 1.5	$15 \times (1.5 \times 2) = 45$

Course Contents:

Foundations - Characterization of DS, System Models, Networking and Inter-networking, Inter-process Communication, Remote Invocation, Indirect Communication and Operating System Support Middle-ware - Dist. Objects and Components, Web Services and Peer- to-Peer Systems System services – Security, Distributed File Systems and Name Services Distributed algorithms - Time and Global States, Coordination and Agreement Shared data, Transactions

and Concurrency Control, Distributed Transactions, and Replication, New challenges - Mobile and Ubiquitous Computing

References:

1. Distributed Systems: Principles and Paradigms, Tanenbum.

Course No.	Course Title	No. of Credits	Credit Hours
SE 4119	Project	Lab: 3	$15 \times (2 \times 3) = 90$

Introduction of the Course: In this course, the students are required to implement a complete software individually by following all the steps of software development life cycle – requirement analysis, design, implementation, testing, deployment.

Specific Objectives:

- To apply industrial experience gathered from internship
- To apply all the skills accumulated throughout their honors life to full-fledged software
- To demonstrate the ability to follow software development life cycle

Semester 8 (4th year 2nd Semester)

Course No.	Course Title	No. of Credits	Credit Hours
SE 4215	Software Security	Theory: 1.5	$15 \times (1 \times 1.5) = 22.5$

Course Contents: Fundamentals of Computer Security, CIA Principles, Security Services and Mechanisms: User Authentication: Password-based authentication, Token-based authentication, Bio-metric authentication, Remote user authentication; Access Control: Discretionary and Role-based access control, Database Security, Sql Injection Attacks, Buffer Overflow attack, Software security touch points: Code Review, Risk Analysis, Penetration Testing, Abuse Cases, Web-vulnerabilities: Cross-site scripting (XSS), cross-site request forgery (CSRF) attacks, Software-penetration testing, Static code analysis, Coding Errors

References:

1. Computer Security: Principles and Practice, by William Stallings and Larry Brown.
2. Software Security: Building Security IN, Gary McGraw

Course No.	Course Title	No. of Credits	Credit Hours
SE 4203	Software Project Management	Theory: 3	$15 \times (3 \times 1) = 45$

Course Contents:

Introduction to Project Management (PM): Project Management (PM) fundamentals: The PM field and job market – People, Process, Product, Technology, historical background and evolution. Basic PM skills, SPM framework, elements, stakeholders, boundaries, challenges of SPM. **Software Project Planning:** planning objective, project plan, variations, structure of SPM plan, project estimation, estimation methods, models, and decision process. **PM Organization and Scheduling:** WBS, types of WBS, functions, activities, tasks, life cycles, phasing and purpose of phasing, building project schedule, network diagrams: PERT, CPM, Bar charts, Gantt charts, Risk and Change Management. **Software Project Management Techniques:** Use of methodologies, managing risks and issues, managing quality, configuration, change, crisis, documentation, release. **Project Monitoring and Control:** Dimensions of monitoring and control, earned value indicators (BCWS, CV, SV, CPI, SPI), backlog management, dispute and error tracking, RMMM charts. **System Test Process:** Test

specifications, black box and white box testing, Test scripts, unit and integration testing, acceptance test specifications, test tools. **Industry Scenarios:** Domain analysis, business case analysis, dynamicity, success and failure factors, case studies

References:

1. Stellman, Andrew, and Jennifer Greene. *Applied software project management*. "O'Reilly Media, Inc.", 2005
2. Phillips, Joseph. *IT project management: on track from start to finish*. McGraw-Hill, Inc., 2002.
3. Rubin, Kenneth S. *Essential Scrum: A practical guide to the most popular Agile process*. Addison-Wesley, 2012.
4. Software Project Management (5th ed.), Bob Hughes and Mike Cotterell, McGraw Hill, (2009)

Course No.	Course Title	No. of Credits	Credit Hours
SE 4216	Software Metrics	Theory: 1.5	$15 \times (1 \times 1.5) = 22.5$

Course Contents:

Overview and scopes of Software Metrics: measurement basics, understanding, control, and improvement, **The Basics of Measurement:** representation theory, measurement scales and scale types, meaningfulness in measurement, **Goal-based framework for software measurement:** determining what to measure, applying the different framework, validation, **Empirical Investigation:** control variables and study type, study goals and hypothesis testing, threats to validity, planning case studies, **Data collection and reliability measurement:** Classical data analysis techniques, hypothesis testing, sample analysis techniques, an overview of statistical tests, **Measuring Internal Attributes: Size and Structure:** code size, design size, functional size, application of size measurements, **Measuring Cost and Effort:** Cocomo models, function point analysis, **Measuring External Attributes: Quality and Reliability:** modeling software quality, usability measures, maintainability measures, and security measures. Basic reliability theories, parametric reliability growth models.

References:

1. Software Metrics, Norman Fenton and James Bieman.

Course No.	Course Title	No. of Credits	Credit Hours
SE 4201	Project	Lab: 3	$15 \times (2 \times 3) = 90$

Introduction of the Course: In this course, the students are required to implement a complete software individually by following all the steps of the software development life cycle – requirement analysis, design, implementation, testing, deployment.

Specific Objectives:

- To apply industrial experience gathered from internship
- To apply all the skills accumulated throughout their honors life to full-fledged software
- To demonstrate the ability to follow software development life cycle

Prerequisite: N/A

Course No.	Course Title	No. of Credits	Credit Hours
SE 4215L	Software Security Lab	Lab: 1.5	$15 \times (1.5 \times 2) = 45$

Course Contents: Demonstration Buffer Overflow, XSS, CSRF, SQL Injections attacks, Static code analysis.

References:

1. Computer Security: Principles and Practice, by William Stallings and Larry Brown.
2. Software Security: Building Security IN, Gary McGraw.

Course No.	Course Title	No. of Credits	Credit Hours
SE 4216	Software Metrics	Theory: 1.5	$15 \times (2 \times 0.75) = 22.5$

Course Contents: Implementation and relevant lab work of the corresponding theory course.

References:

1. Software Metrics, Norman Fenton and James Bieman.

Elective Courses

Course No.	Course Title	No. of Credits	Credit Hours
CSE 4270	Image Processing	Theory: 3	15 × (3×1) = 45

Course Contents: Digital Image Fundamentals - Elements of Visual Perception, Light and the Electromagnetic Spectrum, Image Sensing and Acquisition, Image Sampling and Quantization, Some Basic Relationships between Pixels, Linear and Nonlinear Operations; Image Enhancement in the Spatial Domain - Background, Basic Gray Level Transformations, Histogram Processing, Enhancement Using Arithmetic/Logic Operations, Basics of Spatial Filtering. Smoothing Spatial Filters, Sharpening Spatial Filters, Combining Spatial Enhancement Methods; Image Enhancement in the Frequency Domain – Background, Introduction to the Fourier Transform and the Frequency Domain, Smoothing Filters, Sharpening Filters; Image Restoration - A Model of the Image Degradation/Restoration Process, Noise Models, Estimating the Degradation Function, Inverse Filtering, Minimum Mean Square Error (Wiener) Filtering, Constrained Least Squares Filtering, Geometric Mean Filter, Geometric Transformations; Image Segmentation- Point, Line, and Edge Detection, K-mean Clustering, Canny Edge Detection, Otsu method; Color Image Processing - Color Fundamentals. Color Models, Pseudo Color Image Processing, Color Transformations, Color Segmentation, Noise in Color Images; Image Compression - Fundamentals. Image Compression Models, Error-Free Compression, Lossy Compression, Image Compression Standards; Morphological Image Processing – Preliminaries, Dilation and Erosion, Opening and Closing, The Hit-or-Miss Transformation, Basic Morphological Algorithms; Image Segmentation - Detection of Discontinuities, Edge Linking and Boundary Detection, Thresholding, Region-Based Segmentation, Boundary Descriptors, Regional Descriptors; Pattern Recognition - Fundamentals of Pattern Recognition, Local and Global features, Hough Transformation, Local Binary Pattern, HoG features, Statistical Pattern Recognition, Principal Component Analysis; Classification and Segmentation- Fundamentals of Classification and Segmentation, Support Vector Machine, Random Forest; Hyperspectral Image Fundamentals; 3D Point Cloud Fundamentals.

References:

1. Digital Image Processing - Rafael C Gonzalez and Richard E. Woods.
2. Pattern Recognition- William Gibson
3. Pattern Classification – R.O. Duda, P.E. Hart, and D.G. Stork

Course No.	Course Title	No. of Credits	Credit Hours
SE 4290	Software Maintenance	Theory: 3	15 × (1×3) = 45

Course Contents:

Software Maintenance Fundamentals: what is software maintenance, why is it important, types of software maintenance; Software Maintenance Measurement: McCabe's Cyclomatic Complexity, Halstead Metrics; Maintenance Process: Maintenance Models, IEEE/EIA 1219 Standard for Maintenance Process, Program Comprehension: Cognition Models for Program Comprehension, Understanding Factors that Affect Program Comprehension, Reengineering: General Model For Software Reengineering, Software Reengineering Strategies, Reverse Engineering: Techniques Used for Reverse Engineering, Change Impact Analysis, Refactoring, DevOps: Towards Easing Software Maintenance: Git Branching Strategies, Observability.

References:

1. Software Maintenance: Concepts and Practice By Penny Grubb, Armstrong A. Takang 2nd edition World Scientific USA.
2. Software Evolution and Maintenance: A Practitioner's Approach by Priyadarshi Tripathy, Kshirasagar Naik (2014, Wiley).

Course No.	Course Title	No. of Credits	Credit Hours
CSE 4250	Natural Language Processing	Theory: 3	15 × (3×1) = 45

Course Contents:

Basic NLP concepts: Regular expressions, tokenization, lemmatization, stemming, edit distance, N-gram language models, parts of speech, named entity; **Review of classification algorithms:** Naive Bayes, Logistic Regression, Neural networks; **Deep neural networks:** CNN, RNN, LSTM; **Vector semantics and embeddings:** Words and vectors, cosine similarity, TF-IDF, word2vec, skip-gram, fastText; **NLP applications:** language generation, machine translation, text summarization, question answering, sentiment analysis.

References:

1. Speech and Language Processing, by Daniel Jurafsky and James H. Martin
2. Natural Language Processing with Python, by Steven Bird and Ewan Klein

Course No.	Course Title	No. of Credits	Credit Hours
CSE 4251	Introduction to Data Mining	Theory: 3	15 × (3×1) = 45

Course Contents:

Introduction to Data mining, Data Warehousing, On-line Analytical Processing; Overview of Clustering, Classification, Rule learning; Data mining process: Data preparation/cleansing and next steps, Association Rule mining, different algorithm types; Classification/Prediction: basics, tree-based approaches, Neural Networks; Clustering: basics, statistical approaches, Neural-net and other approaches; Text Mining, Time Series Mining, Mining Data Streams, Multi-Relational Data Mining, Data Mining for Fraud Detection

References:

1. Ian H. Witten and Eibe Frank, Data Mining: Practical Machine Learning Tools and Techniques (Second Edition), Morgan Kaufmann, 2005, ISBN: 0-12-088407-0.
2. Data Mining: Concepts and Techniques (The Morgan Kaufmann Series in Data Management Systems) 3rd Edition

Course No.	Course Title	No. of Credits	Credit Hours
CSE 4271	Photogrammetry	Theory: 3	15 × (3×1) = 45

Course Contents:

Introduction to Photogrammetry, Geometry of a single image, Characteristics of vertical imagery, Stereophotogrammetry, Aerial cameras and Photographs, Geometry of Aerial Photograph, Introduction to Ortho-photos and DEM/contour extraction, Aerial Photo Interpretation techniques and tools, Mapping from aerial photographs, Introduction to laser scanning, LiDAR data processing, classification, and segmentation.

Reference Books and Journals:

1. Introduction to Remote Sensing, Fifth Edition – James B. Campbell.
2. ISPRS Journal of Photogrammetry and Remote Sensing

3. International Journal of Remote Sensing
4. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing

Course No.	Course Title	No. of Credits	Credit Hours
CSE 4280	Introduction to Bioinformatics	Theory: 3	15 × (1×3) = 45

Course Contents: Essentials of Molecular biology: DNA, RNA and Protein, Watson and Crick Model of DNA, DNA replication, transcription, translation, splicing, Central dogma of molecular biology; DNA sequencing technology; Sequence Databases; Sequence Formats, Single-cell RNA-sequences, microarrays; Pairwise sequence alignment: local and global alignment, amino acid substitution scoring matrices; significance of the alignment, Multiple sequence alignment: progressive, iterative, statistical methods for multiple sequence alignment; local multiple alignment - sequence profiles and motifs, EM algorithm, MCMC algorithm, Gibbs Algorithm, Variational Inference; Hidden Markov Models: theory; training and applications to sequence alignment. Sequence database search: heuristic methods – FASTA, BLAST; Global Alignment methods – MatGat, Multiple Alignment using Clustal; Basic methods in molecular phylogeny: phylogenetic trees; distance matrix methods; maximum parsimony methods; maximum likelihood methods. RNA secondary structure basics: energy minimization, comparative sequence analysis, combined sequence and computational method; Gene Prediction: neural network, pattern discrimination functions and HMM; Promoter prediction: pattern driven, sequence driven algorithms; Proteins: protein classification, structure alignment and prediction; microarrays: design, data acquisition and analysis; gene network modeling; Systems biology.

References:

1. Bioinformatics: Sequence and Genome Analysis by David W. Mount, Cold Spring Harbor Laboratory Press; 2001.
2. Bioinformatics: the machine learning approach by Baldi P, Brunak S., MIT press; 2001.
3. An introduction to bioinformatics algorithms by Jones, N.C. and Pevzner, P.A., MIT press; 2004.
4. Bioinformatics: genes, proteins and computers by Orengo, C., Jones, D. and Thornton, J. eds., Taylor & Francis; 2003.
5. Next-Generation Sequencing Data Analysis by Xinkun Wang, CRC Press; 2016.

Course No.	Course Title	No. of Credits	Credit Hours
CSE 4281	Genomics Data Science	Theory: 3	15 × (1×3) = 45

Course Contents: Introduction to the statistics for genomics data science: Data normalization, exploratory analysis, linear modeling, testing, and multiple testing; preprocessing, batch effects, hypothesis testing, multiple hypothesis testing; Generative model for non-continuous outcomes – binary data, count data; mixture models, clustering, visualization; Multivariate analysis – high throughput genomics data, Multivariate methods for heterogeneous data, supervised learning, networks and trees. Scoring matrices: Protein and nucleotide scoring matrices i.e.PAM, BLOSUM, Gonett. How to construct scoring matrices. Difference between PAM and Blosun. Database homology search: Concepts behind BLAST: Applications & Biological Significance; homology, similarity & identity Statistical significance of BLAST: E value, Scores BLAST versions- BLASTp, BLASTn, Difference between FASTA and BLAST. Single-cell genomics analysis: Quality control, preprocessing and normalization, Exploratory analysis – Cell type annotation and identification, trajectory inference and analysis. Phylogenetic analysis: Basic terminology in Phylogenetics: Distance and parsimony methods; Clustering methods. Rooted and un-rooted trees. Predictive methods using DNA sequences: Gene predictive methods- searching by signal, searching by content, homology based predictions, Markov models, Hidden Markov models in gene prediction: Genscan, Glimmer, Grail. Promoter analysis and predictions. Protein Structure Prediction: Secondary structure prediction methods: CHAU FASMAN, GOR, NN Tertiary Structure prediction methods- Homology Modeling, Threading/Fold recognition and Ab initio.

References:

1. Modern Statistics for Modern Biology by Susan Holmes and Wolfgang Huber, Cambridge University Press; 2019.
2. Bioinformatics: Sequence and Genome Analysis by David W. Mount, Cold Spring Harbor Laboratory Press; 2001.
3. Bioinformatics: the machine learning approach by Baldi P, Brunak S., MIT press; 2001.
4. An introduction to bioinformatics algorithms by Jones, N.C. and Pevzner, P.A., MIT press; 2004.
5. Next-Generation Sequencing Data Analysis by Xinkun Wang, CRC Press; 2016.
6. Computational Genome Analysis: An Introduction (Statistics for Biology & Health S) by Richard C. Deonier, Simon Tavaré and Michael S. Waterman, Springer; 2005.